University of Exeter

MTH1003

Mathematical Modelling

On Campus Exam	**Closed Book**
Additional Materials: None	**Permitted Materials:** Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

-The **duration** of this paper is 2 hours
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

SECTION A

1. (a) A particle of mass $m = 2$ is initially at rest at position $-\boldsymbol{j} - \alpha\boldsymbol{k}$ where α is a real-valued constant. The particle is acted on by a time dependent force

$$\boldsymbol{F}(t) = \cos t\, \boldsymbol{i} + \frac{4}{(t-1)^3}\, \boldsymbol{j}.$$

Find the velocity $\boldsymbol{v}$ and position $\boldsymbol{r}$ of the particle as functions of time and determine the value of α for which the position vector $\boldsymbol{r}$ is perpendicular to the vector $\boldsymbol{i} + \boldsymbol{k}$ at $t = \pi/2$. (8)

(b) Write down the Lagrange polynomials $\mathcal{L}_k(x)$ for $k = 1, 2, 3$ using the three points $x_1 = 0$, $x_2 = \pi/4$, $x_3 = \pi/2$ and hence write down the quadratic function $P(x)$ that interpolates the function $f(x) = \sin x + \cos x$ at those points. Evaluate $P(x)$ at $x = \pi/6$ and compare to the true value of $f(x)$ at this point. (8)

(c) (i) State Hooke's law for the tension in an elastic string, defining any mathematical variables that you introduce. (4)

(ii) A particle of mass m is attached to one end of a light elastic string and the other end is fixed to a rigid support. The string hangs vertically under gravity, where g is the gravitational acceleration. Obtain the equation of motion for vertical displacements of the particle in terms of its distance x from the support. Find the equilibrium position of the particle. (6)

(iii) When in its equilibrium position the particle is given an upward velocity U; under what condition on U does the string become slack during the subsequent motion? (4)

(d) A particle with position $x(t)$ undergoes damped harmonic motion governed by

$$\ddot{x} + 2k\dot{x} + \omega^2 x = 0$$

with initial conditions $x = 3$, $\dot{x} = 0$ at $t = 0$. For the two cases

(i) $k = 2, \quad \omega^2 = 4,$ (6)

(ii) $k = 1, \quad \omega^2 = 10,$ (6)

obtain the general solution and state what kind of damping occurs.

(e) You are **given** that the Lorentz transformation between coordinates (x, t) for an observer $\mathcal{O}$ and coordinates (x', t') for

an observer $\mathcal{O}'$ is

$$x' = \gamma\big(x - ut\big), \quad t' = \gamma\big(t - ux/c^2\big), \quad \gamma = \frac{1}{\sqrt{1 - u^2/c^2}}\,,$$

where u is the velocity of $\mathcal{O}'$ with respect to $\mathcal{O}$ and c is the speed of light.

(i) Show that the space–time interval between $(0, 0)$ and (x, t), defined by

$$s^2 = c^2t^2 - x^2,$$

is the same for both observers, $\mathcal{O}$ and $\mathcal{O}'$. In other words show that $c^2t'^2 - x'^2 = s^2$. (5)

(ii) Hence show that if photons leave $(0, 0)$ travelling at the speed of light as measured by $\mathcal{O}$, then the same photons are observed by $\mathcal{O}'$ also to travel at the speed of light. (3)

[**50**]

SECTION B

2. A ball is launched from the origin on flat ground in the $x - z$ plane with initial speed V at an angle α to the horizontal. Let g be the gravitational acceleration.

   (a) Assuming the ball meets no obstacle, and modelling it as a particle, write down its x and z coordinates as functions of time. Hence obtain the trajectory equation of the ball. (5)

   (b) Now suppose that there is a wall of height H at position $x = 3H$ and that the launch speed of the ball is $V = 3\sqrt{\frac{gH}{2}}$. Determine the range of angles α for which the ball goes over the wall. If the ball is launched at an angle $\alpha = \pi/3$, where will it hit the ground? (10)

   (c) Now suppose that the ball is partially elastic. It is launched from the origin at an angle $\alpha = \pi/4$, with velocity as given above, that is $V = 3\sqrt{\frac{gH}{2}}$. Where will the ball hit the ground? Assume that when it hits the ground it bounces at the same angle but loses half of its energy. What is the maximum possible height of a wall placed at $x = 13H/2$ that will not obstruct the ball's motion? (10)

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3. A general Lotka–Volterra system is given by the equations

$$\dot{H} = -sH + bHP, \tag{1}$$

$$\dot{P} = rP(1 - qP) - aHP, \tag{2}$$

where $P(t) \geq 0$ gives the population of prey and $H(t) \geq 0$ the population of predators (hunters), in some units. Here a, b, q, r and s are parameters that are positive or zero.

(a) In the case $a = b = 1$, $q = 1$, $s = 1/3$, $r = 1$, find and sketch the nullclines. Classify the equilibrium points and hence sketch the possible behaviour of trajectories in the (H, P) phase space. (15)

(b) For the general system given by equations (1) and (2), say briefly what each of the parameters (a, b, q, r and s) means in the context of a population of predators $H(t)$ interacting with a population of prey $P(t)$. (5)

(c) In the case $q = 0$ but all other parameters taking general positive values, apply separation of variables to the equation for dH/dP (or use another method) to show that a conserved quantity for the system given by (1) and (2) is

$$f(H, P) = H^{-r}\, P^{-s}\, e^{aH+bP}\,.$$

(5)

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4. (a) (i) Write down the Taylor series expansions about the point x for $f(x+h)$, $f(x+2h)$ and $f(x+3h)$, up to and including terms containing third derivatives. Hence deduce the order of the finite-difference approximation to $f'(x)$ given by:

$$f'(x) \approx \frac{-11f(x) + 18f(x+h) - 9f(x+2h) + 2f(x+3h)}{6h}.$$

(6)

(ii) The python code below uses this approximation. Find the two errors in the code, describe how to fix them, and describe what the fixed code does.

```
import numpy as np
import matplotlib.pyplot as plt

err = []
hvals = []

for p in range(1, 8):
    h = 0.5**p
    hvals.append(h)
    dfdx = (-11*np.sin(1) + 18*np.sin(1+h) \
            -9*np.sin(1+2h) + 2*np.sin(1+3*h))/6*h
    err.append(abs(np.cos(1)-dfdx))

plt.loglog(hvals, err, 'ko')
plt.show()
```

(6)

(b) You are **given** the equation of motion in polar coordinates (r, θ) for a particle of mass m moving under a central force $\boldsymbol{F}(r) = mf(r)\hat{\boldsymbol{r}}$ as:

$$(\ddot{r} - r\dot{\theta}^2)\hat{\boldsymbol{r}} + \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)\hat{\boldsymbol{\theta}} = f(r)\hat{\boldsymbol{r}},$$

where $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\theta}}$ are unit vectors in the radial and azimuthal directions respectively.

(i) Define the angular momentum h per unit mass and show that it is conserved.

(ii) Eliminate $\dot{\theta}$ in terms of h and r, and hence re-express the radial component of the equation of motion in terms of $u = 1/r$ with independent variable θ.

(iii) Give the equation for u that results in the case of an inverse cube law $f(r) = -\mu r^{-3}$. In the case when $h^2 < \mu$ write down the general solution $u(\theta)$ and comment on the behaviour of the particle.

(13)

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